



Develop a Solution Lab

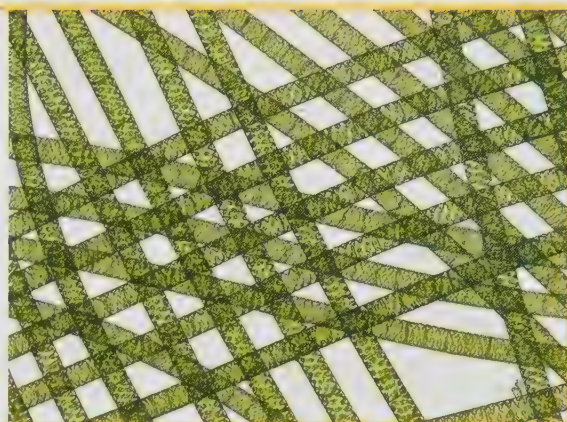
Open-ended Inquiry

Algae in the Water

Problem What are the ideal growth conditions for algae?

In this lab, you will plan and conduct an experiment to determine the effects of a variable, such as temperature or sunlight, on algae growth. Then you will propose ideas for a system for raising algae.

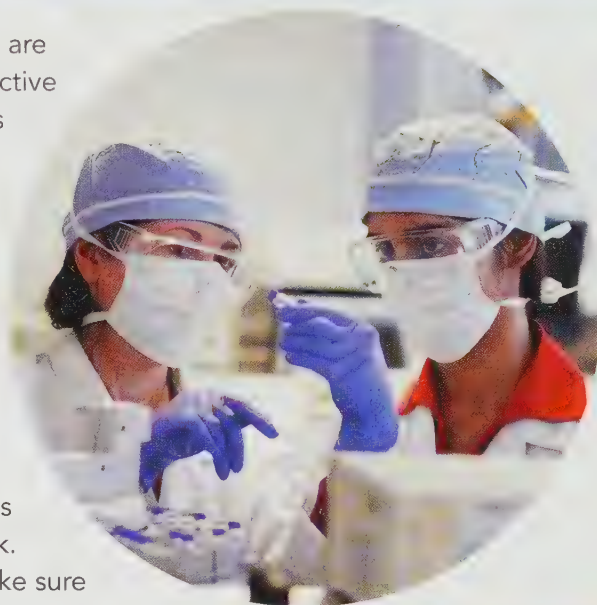
You can find this lab in your digital course.



Safety Scientists, such as those shown in **Figure 1-14**, are trained to use safety procedures, such as wearing protective glasses and gloves. Laboratory work may involve flames or heating elements, electricity, chemicals, hot liquids, sharp instruments, or breakable glassware. Laboratory work and fieldwork may involve contact with living or dead organisms—not just potentially poisonous plants and venomous animals but also disease-carrying organisms and substances contaminated with dangerous microorganisms.

Whenever you work in your biology laboratory, you must follow safe practices. Careful preparation is the key to staying safe. Before performing any activity in this course, study the safety rules in the Lab Skills Handbook. Before you start each activity, read all the steps and make sure you understand them, including safety precautions.

The single most important safety rule is to always follow your teacher's instructions and directions in this textbook. Anytime you are in doubt, ask your teacher for an explanation. And because you may come in contact with organisms you cannot see, it is essential that you wash your hands thoroughly with soap and warm water after every scientific activity. Remember that you are responsible for your own safety and that of your teacher and classmates. If you are handling live animals, you are responsible for their safety, too.



INTERACTIVITY

Figure 1-14

Safety in the Laboratory

These scientists are wearing protective glasses and gloves in keeping with the safety protocols in their lab.



LESSON 1.3 Review

KEY QUESTIONS

1. How do living things differ from nonliving things?
2. How do crosscutting concepts help unite the study of biology?
3. How can biology be studied at different scales, or levels?
4. Why do scientists use a common system of measurement?

CRITICAL THINKING

5. **Identify Patterns** To analyze the characteristics of life, why is it useful to identify patterns at the level of molecules and cells?
6. **Calculate** In an experiment, you need 250 grams of potting soil for each of 10 plant samples. How many kilograms of soil in total do you need?